Assignment #6: Isotope Effects

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The Morita-Baylis-Hillman coupling reaction is a popular approach to forming a carbon-carbon bond between the α-carbon of an unsaturated ester and an aldehyde group, facilitated by a nucleophilic catalyst. It will be the subject of this week's assignment.

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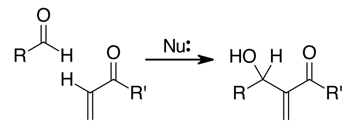

Isotope Effects Reveal Mechanisms

The Morita-Baylis-Hillman reaction has been the focus of many investigations into its mechanism.[1] In this assignment, we will consider a communication from Tyler McQuade's group at Cornell University.[2] In this work, a series of experiments using kinetic isotope effects provide evidence for identifying the rate-determining step in the mechanism.

The Activity

Read the communication from McQuade et al. and seek to understand the mechanism as proposed and the evidence for that mechanism.[3]

The Report

Write a report in which you provide your interpretation of the mechanism proposed by the authors and discuss how the evidence presented in this communication is consistent with the hypothesis. Be sure to describe both isotopic substitution experiments in your discussion and explain the values of the kinetic isotope effects.

The Rubric

I am looking for the following elements as I grade the report:

- A diagram that presents the mechanism with all important steps included. Use curved arrows to show electron flow. Describe the major steps.[4]
- A discussion of the kinetic experiments and how the results support the proposed mechanism (or not).
- A discussion of the two kinetic isotope effect experiments with an explanation for the magnitudes of the observed effects and how the observations support the proposed mechanism (or not).
- Include any appropriate references.[5]
- Quality original writing.

[1] Whenever you encounter a reaction that you don't know, Google is your friend. I encountered this reaction while reading (try to read one synthetic paper each week to exercise your organic chemistry mechanism skills) and a quick search led me to a very well written Wikipedia article at https://en.wikipedia.org/wiki/Baylis-Hillman_reaction. The paper that is featured in this assignment was referenced on that page.

[2] Dr. McQuade began his academic career at Cornell, moved to Florida State in 2007 and to Virginia Commonwealth University in 2017. He is now a corporate executive at On Demand Pharmaceuticals, a company much like our own BioVectra.

[3] "Baylis-Hillman Mechanism: A New Interpretation in Aprotic Solvents." K.E. Price, S.J. Broadwater, H.M. Jung, D.T. McQuade, *Org. Lett.*, **2005**, *7*, 147-150. https://doi.org/10.1021/ol047739o

[4] I am looking for a written description of the mechanism diagram. The authors of the paper provide a lot of description that you can consider as you choose your own words.

[5] Avoid websites. Sometimes we need to go there, but usually, there are better sources in the published literature. Remember, the textbook is a legitimate source.

NOTE: The paper for this assignment is a communication. It is a short note to let the world know about some awesome results and implies that the authors will publish the full story eventually. You could compare the communication with the full paper that was published six months later. See "A New Interpretation of the Baylis-Hillman Mechanism." K.E. Price, S.J. Broadwater, B.J. Walker, D.T. McQuade, *J. Org. Chem.*, **2005**, *70*, 3980-3987. https://doi.org/10.1021/jo050202j. For fun, you could check out the 150 pages of supplementary material. Now there's some data that we could dig into on another day.